

Homework 1

Incentive-Centered Design

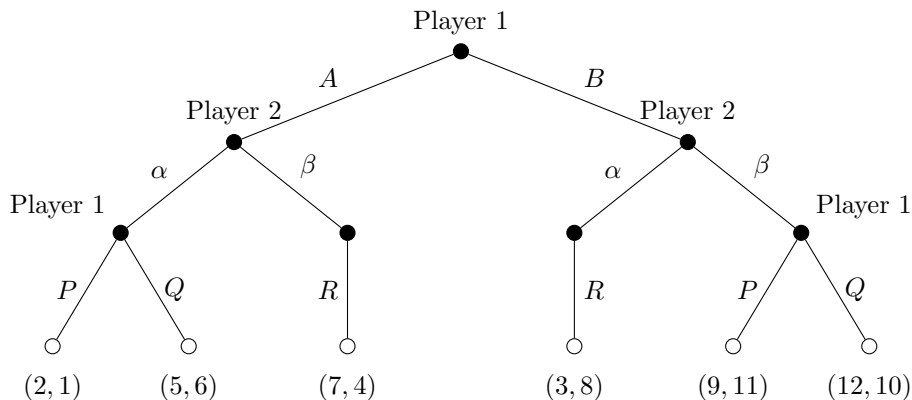
Jan-May '26

Acknowledgements

1. Collaborators: Anshuman Mishra (ED23B006) - Individually completed
2. LLMs referred:
ChatGPT - LaTeX formatting and tikz diagrams, understanding potential games better and clarifying generic doubts.
Gemini - Used for getting deeper clarity and ideas for the proofs. Also helped with notations slightly.

Solutions

1. Consider a 2-player extensive form game specified by the following Tree. At time $t = 1$ and $t = 3$, it is Player 1's turn to play, and at time $t = 2$, it is Player 2's turn to play. The available actions at each time are specified in the Tree, and both players know the Tree.



- (a) What action do you think Player 1 will play at time 1, Player 2 will play at time 2, and Player 3 will play at time 3?
- (b) Identify the set of strategies available to each player.
- (c) Express the game in strategic form.
- (d) Identify all pure-strategy Nash equilibria of the game.
- (e) Is the action sequence identified in Part (a) generated by any Nash equilibrium? Are any other action sequences also generated by a Nash equilibrium?
- (f) In the presence of a multiplicity of Nash equilibria, refined notions of equilibria can be defined to weed out certain Nash equilibria. One such notion for extensive-form games is *subgame-perfect equilibrium*.

A subgame of an extensive-form game is a game defined by any sub-Tree of the game. For example, in the above game, one can begin observing the game from $t = 2$ so that the first move is played by Player 2 and the second move is played by Player 1. There are two such 2-round sub-games in the above game — one corresponding to each node in the second layer. Similarly, there are four 1-round games. The above game, therefore, has seven sub-games (the original game is also a subgame).

A Nash equilibrium of an extensive-form game is said to be a subgame-perfect equilibrium if the strategy profile, when implemented on any subgame, also forms a Nash equilibrium. Read up more on subgame-perfect equilibrium from your favorite source.

Which of the Nash equilibria identified in Part (d) are subgame-perfect equilibria? Do the action sequences generated by subgame-perfect equilibria coincide with the action sequence you identified in Part (a)?

Solution:**(a) Solution using backward induction**

We can solve the game by backward induction i.e working our way up from the last performed move.

Step 1: Player 1's optimal actions at the final stage ($t = 3$).

- After (A, α) : Player 1 compares payoffs 2 (action P) and 5 (action Q), and chooses Q .
- After (A, β) : Player 1 chooses R (only available action).
- After (B, α) : Player 1 chooses R (only available action).
- After (B, β) : Player 1 compares payoffs 9 (action P) and 12 (action Q), and chooses Q .

Step 2: Player 2's optimal actions at $t = 2$.

- If Player 1 plays A :
 - * $\alpha \rightarrow$ outcome $(5, 6)$
 - * $\beta \rightarrow$ outcome $(7, 4)$
 Player 2 prefers α since $6 > 4$.
- If Player 1 plays B :
 - * $\alpha \rightarrow$ outcome $(3, 8)$
 - * $\beta \rightarrow$ outcome $(12, 10)$
 Player 2 prefers β since $10 > 8$.

Step 3: Player 1's optimal action at $t = 1$.

- If Player 1 plays A , the outcome is $(5, 6)$.
- If Player 1 plays B , the outcome is $(12, 10)$.

Player 1 prefers B since $12 > 5$.

Conclusion: The outcome induced by backward induction is $[(B, \beta, Q)]$.

(b) Strategy sets

A strategy for a player specifies an action at every decision node of that player.

Player 1: Player 1 moves at $t = 1$, and again at $t = 3$ at four possible nodes: (A, α) , (A, β) , (B, α) , and (B, β) . However, at (A, β) and (B, α) , the action is uniquely R .

Thus, Player 1 only has choices at:

* the initial node: A or B , * (A, α) : P or Q , * (B, β) : P or Q .

Hence, Player 1 has $2 \times 2 \times 2 = 8$ strategies, which we write in the form (initial, action at (A, α) , action at (B, β)):

$$[S_1 = APP, APQ, AQP, AQQ, BPP, BPQ, BQP, BQQ.]$$

Player 2: Player 2 moves at two nodes: after A and after B . Hence, a strategy must specify both actions:

$$[S_2 = (\alpha, \alpha), (\alpha, \beta), (\beta, \alpha), (\beta, \beta).]$$

Here, the first component corresponds to the action after A , and the second to the action after B .

(c) Strategic (normal) form representation

The payoff for each strategy profile is determined by following the path induced by the strategies.

Observations:

- If Player 1 plays A , the outcome depends only on Player 2's first component.
- If Player 1 plays B , the outcome depends only on Player 2's second component.

	(α, α)	(α, β)	(β, α)	(β, β)
APP	(2, 1)	(2, 1)	(7, 4)	(7, 4)
APQ	(2, 1)	(2, 1)	(7, 4)	(7, 4)
AQP	(5, 6)	(5, 6)	(7, 4)	(7, 4)
AQQ	(5, 6)	(5, 6)	(7, 4)	(7, 4)
BPP	(3, 8)	(9, 11)	(3, 8)	(9, 11)
BPQ	(3, 8)	(12, 10)	(3, 8)	(12, 10)
BQP	(3, 8)	(9, 11)	(3, 8)	(9, 11)
BQQ	(3, 8)	(12, 10)	(3, 8)	(12, 10)

To identify Nash equilibria, we mark best responses:

- For Player 1: in each column, select the strategy that maximizes Player 1's payoff.
- For Player 2: in each row, select the strategy that maximizes Player 2's payoff.

The intersections of these best responses correspond to pure-strategy Nash equilibria.

(d) Pure-strategy Nash equilibria

A strategy profile is a Nash equilibrium if each player's strategy is a best response to the other's. We identify these by marking Player 1's best responses (column-wise maxima) and Player 2's best responses (row-wise maxima). The intersections are highlighted in red. These points correspond to Pure-strategy Nash Equilibria

	(α, α)	(α, β)	(β, α)	(β, β)
<i>APP</i>	(2, 1)	(2, 1)	(7, 4)	(7, 4)
<i>APQ</i>	(2, 1)	(2, 1)	(7, 4)	(7, 4)
<i>AQP</i>	(5, 6)	(5, 6)	(7, 4)	(7, 4)
<i>AQQ</i>	(5, 6)	(5, 6)	(7, 4)	(7, 4)
<i>BPP</i>	(3, 8)	(9, 11)	(3, 8)	(9, 11)
<i>BPQ</i>	(3, 8)	(12, 10)	(3, 8)	(12, 10)
<i>BQP</i>	(3, 8)	(9, 11)	(3, 8)	(9, 11)
<i>BQQ</i>	(3, 8)	(12, 10)	(3, 8)	(12, 10)

Pure strategies corresponding to nash equilibria:

$$\begin{aligned}
& (APP, (\beta, \alpha)), (APQ, (\beta, \alpha)), \\
& (AQP, (\alpha, \alpha)), (AQQ, (\alpha, \alpha)), \\
& (BPQ, (\alpha, \beta)), (BQQ, (\alpha, \beta)), \\
& (BPQ, (\beta, \beta)), (BQQ, (\beta, \beta)).
\end{aligned}$$

Mixed strategies: Since multiple pure-strategy Nash equilibria exist, there is no need to compute mixed-strategy equilibria. Mixed equilibria may exist due to indifference, but they are not required here.

- (e) Yes, the action sequence (B, β, Q) identified in part (a) is generated by Nash equilibria such as $(BPQ, (\beta, \beta))$ and $(BQQ, (\beta, \beta))$.

Other action sequences generated by Nash equilibria include:

- (A, α, Q) arising from $(AQP, (\alpha, \alpha))$ and $(AQQ, (\alpha, \alpha))$,
- (A, β, R) arising from $(APP, (\beta, \alpha))$ and $(APQ, (\beta, \alpha))$.

- (f) A subgame-perfect equilibrium (SPE) is a strategy profile that induces a Nash equilibrium in every subgame.

From backward induction, the unique SPE is:

$$(B, \beta, Q).$$

All other Nash equilibria fail to be subgame perfect because they prescribe actions that are not optimal in some proper subgames. In particular, they rely on non-credible threats, i.e., actions that would not be chosen if the corresponding subgame were actually reached. Thus, only (B, β, Q) survives as a subgame-perfect equilibrium. Therefore, the action sequence generated by the SPE coincides with the one identified in part (a).

2. This question asks you to prove the proposition in Lecture 4, Slide 15. Let $\mathcal{X} \subseteq \mathbb{R}^m$ and $\mathcal{Y} \subseteq \mathbb{R}^n$ be any two sets, and let $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be any function. Show that there exists $(\mathbf{x}^*, \mathbf{y}^*)$ satisfying

$$\mathbf{x}^* = \arg \max_{\mathbf{q}} f(\mathbf{q}, \mathbf{y}^*), \quad (1)$$

$$\mathbf{y}^* = \arg \min_{\mathbf{q}} f(\mathbf{x}^*, \mathbf{q}) \quad (2)$$

if and only if

$$\max_x \min_y f(x, y) = \min_y \max_x f(x, y). \quad (3)$$

Solution: Solution to Question 2

Let $X \subseteq \mathbb{R}^m$, $Y \subseteq \mathbb{R}^n$, and let $f : X \times Y \rightarrow \mathbb{R}$.

Suppose there exists (x^*, y^*) such that:

$$x^* \in \arg \max_q f(q, y^*), \quad y^* \in \arg \min_q f(x^*, q).$$

We can define two functions as

$$g(x) := \min_y f(x, y), \quad h(y) := \max_x f(x, y).$$

By definition of minimum and maximum, we have

$$g(x) \leq f(x, y) \leq h(y) \quad \forall x \in X, \forall y \in Y.$$

Taking maximum over x in the left inequality,

$$\max_x g(x) \leq f(x, y) \quad \forall y.$$

Now taking minimum over y ,

$$\max_x g(x) \leq \min_y h(y).$$

Thus,

$$\max_x \min_y f(x, y) \leq \min_y \max_x f(x, y).$$

Now, since x^* and y^* satisfy

$$f(x, y^*) \leq f(x^*, y^*) \leq f(x^*, y) \quad \forall x, y,$$

we obtain

$$\max_x f(x, y^*) = f(x^*, y^*), \quad \min_y f(x^*, y) = f(x^*, y^*).$$

Substituting into the inequality we get,

$$\max_x \min_y f(x, y) \leq \min_y f(x^*, y) = f(x^*, y^*) = \max_x f(x, y^*) \leq \min_y \max_x f(x, y).$$

Hence,

$$\max_x \min_y f(x, y) = \min_y \max_x f(x, y).$$

Conversely, suppose that

$$\max_x \min_y f(x, y) = \min_y \max_x f(x, y) = v.$$

Let $x^* \in \arg \max_x \min_y f(x, y)$ and $y^* \in \arg \min_y \max_x f(x, y)$.

Then,

$$\min_y f(x^*, y) = v \quad \text{and} \quad \max_x f(x, y^*) = v.$$

From the first,

$$f(x^*, y) \geq v \quad \forall y,$$

and from the second,

$$f(x, y^*) \leq v \quad \forall x.$$

Combining,

$$f(x, y^*) \leq v \leq f(x^*, y) \quad \forall x, y.$$

In particular,

$$f(x^*, y^*) = v,$$

and hence,

$$f(x, y^*) \leq f(x^*, y^*) \leq f(x^*, y) \quad \forall x, y.$$

Thus,

$$x^* \in \arg \max_x f(x, y^*), \quad y^* \in \arg \min_y f(x^*, y),$$

which establishes the existence of (x^*, y^*) .

3. Consider a two-player zero-sum game with the maximizer having m pure strategies and the minimizer having n pure strategies. Denote by $\mathbf{x} \in \mathbb{R}^m$ the mixed strategy of the maximizer, by $\mathbf{y} \in \mathbb{R}^n$ the mixed strategy of the minimizer, and by $\mu(\mathbf{x}, \mathbf{y})$ the expected payoff of the maximizer. Show that $\mu : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ is concave-convex, i.e., $\mu(\cdot, \mathbf{y})$ is concave for each $\mathbf{y}$ and $\mu(\mathbf{x}, \cdot)$ is convex for each $\mathbf{x}$.

Solution:

Let $A = (A_{ij}) \in \mathbb{R}^{m \times n}$ be the payoff matrix, where A_{ij} denotes the payoff when the maximizer plays pure strategy i and the minimizer plays pure strategy j .

Let $x = (x_1, \dots, x_m)$ and $y = (y_1, \dots, y_n)$ be mixed strategies. Then the expected payoff is obtained by taking expectation over all pure strategy profiles:

$$\mu(x, y) = \sum_{i=1}^m \sum_{j=1}^n x_i y_j A_{ij}.$$

since $x_i y_j$ is the probability that the strategy profile (i, j) is played. This can be written in matrix form as

$$\mu(x, y) = x^\top A y.$$

Concavity in x :

Fix y . Let $x_1, x_2, \dots, x_k \in \mathbb{R}^m$ be pure strategies and let $\lambda_1, \dots, \lambda_k \geq 0$ such that

$$\sum_{i=1}^k \lambda_i = 1.$$

Thus x can be described as a convex combination given by,

$$x = \sum_{i=1}^k \lambda_i x_i.$$

Then,

$$\mu(x, y) = \left(\sum_{i=1}^k \lambda_i x_i \right)^\top A y.$$

Using linearity,

$$= \sum_{i=1}^k \lambda_i x_i^\top A y = \sum_{i=1}^k \lambda_i \mu(x_i, y).$$

Thus,

$$\mu\left(\sum_{i=1}^k \lambda_i x_i, y\right) = \sum_{i=1}^k \lambda_i \mu(x_i, y),$$

which shows that $\mu(\cdot, y)$ is linear (hence concave).

Convexity in y :

Fix x . Let $y_1, y_2, \dots, y_k \in \mathbb{R}^n$ and $\lambda_1, \dots, \lambda_k \geq 0$ with

$$\sum_{i=1}^k \lambda_i = 1.$$

Let

$$y = \sum_{i=1}^k \lambda_i y_i.$$

Then,

$$\mu(x, y) = x^\top A \left(\sum_{i=1}^k \lambda_i y_i \right).$$

Using linearity,

$$= \sum_{i=1}^k \lambda_i x^\top A y_i = \sum_{i=1}^k \lambda_i \mu(x, y_i).$$

Thus,

$$\mu\left(x, \sum_{i=1}^k \lambda_i y_i\right) = \sum_{i=1}^k \lambda_i \mu(x, y_i),$$

which shows that $\mu(x, \cdot)$ is linear (hence convex)

Thus, $\mu(x, y)$ is linear in each argument separately. Hence, It is concave in x and convex in y , making it concave-convex.

4. Consider a 2-player normal form game in which the row player has m pure strategies and the column player has n pure strategies. Suppose that no two strategy profiles have the same payoff for any player. What is maximum number of Pure-strategy Nash Equilibria (PNE) that the game can have?

If not for the constraint that all payoffs are distinct, what is the maximum number of PNE that the game can have?

Solution:

A pure-strategy Nash equilibrium (PNE) is basically a strategy profile that is simultaneously:

- a best response of the row player in its column, and
- a best response of the column player in its row.

Case 1: Distinct payoffs

If all payoffs are distinct for each player, then:

- In every column, the row player has a *unique* best response.
- In every row, the column player has a *unique* best response.

Thus, there are exactly n row best-response marks (one per column) and m column best-response marks (one per row). A PNE occurs at an intersection of these marks, so the total number of PNE is at most

$$\min(m, n).$$

This bound is tight. Let $k = \min(m, n)$ and select k cells such that no two lie in the same row or column. Assign payoffs so that in each selected cell, both players obtain strictly higher payoffs than in any other cell in the same row or column (while keeping all payoffs distinct). Each such cell becomes a mutual best response, yielding k Nash equilibria. Hence,

max number of PNE with distinct payoffs = $\min(m, n)$.
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Example (distinct payoffs, $m = 3, n = 4$):

	C_1	C_2	C_3	C_4
R_1	(8, 8)	(1, 2)	(0, 1)	(2, 3)
R_2	(2, 1)	(9, 9)	(3, 0)	(1, 4)
R_3	(1, 0)	(2, 3)	(10, 10)	(0, 2)

Red entries denote pure-strategy Nash equilibria. There are $3 = \min(3, 4)$ PNE.

Case 2: Without distinctness

If ties are allowed, a player may have multiple best responses. In particular, it is possible that:

- in every row, all columns are best responses for the column player, and
- in every column, all rows are best responses for the row player.

In this case, every cell is a mutual best response, and hence every profile is a PNE. Therefore,

max number of PNE without distinctness = mn .

Example (ties allowed, $m = n = 3$):

	C_1	C_2	C_3	C_4
R_1	(1, 1)	(1, 1)	(1, 1)	(1, 1)
R_2	(1, 1)	(1, 1)	(1, 1)	(1, 1)
R_3	(1, 1)	(1, 1)	(1, 1)	(1, 1)

All entries are pure-strategy Nash equilibria, so there are $12 = mn$ PNE.

5. Identify all pure-strategy Nash equilibria and mixed-strategy Nash equilibria of the game with the payoff matrix

$$\begin{bmatrix} (1, 2) & (1, 3) \\ (0, 3) & (4, 1) \end{bmatrix}. \quad (4)$$

Solution:

	C_1	C_2
R_1	$(1, 2) \xrightarrow{\text{red}} (1, 3)$	
R_2	$(0, 3) \xleftarrow{\text{red}} (4, 1)$	

Since it is apparent that no position in the payoff matrix corresponds to a pure-strategy Nash equilibrium, no pure Nash equilibrium exists. This is evident from the **blue arrows**, which indicate profitable deviations for the row player, and the **red arrows**, which indicate profitable deviations for the column player. There is no cell which is a mutual best response, as in every outcome, at least one player has a profitable deviation.

We therefore proceed to compute a mixed-strategy Nash equilibrium by letting the row player play $(p, 1 - p)$ and the column player play $(q, 1 - q)$.

Row player: $(p, 1 - p)$ over (R_1, R_2)

Column player: $(q, 1 - q)$ over (C_1, C_2)

At a mixed Nash equilibrium, the row player must be indifferent between strategies in the support,

$$\text{Payoff}(R_1) = q(1) + (1 - q)(1) = 1$$

$$\text{Payoff}(R_2) = q(0) + (1 - q)(4) = 4(1 - q)$$

$$1 = 4(1 - q) \Rightarrow q = \frac{3}{4}$$

Similarly, the column player must be indifferent between strategies in the support,

$$\text{Payoff}(C_1) = p(2) + (1 - p)(3) = 3 - p$$

$$\text{Payoff}(C_2) = p(3) + (1 - p)(1) = 1 + 2p$$

$$3 - p = 1 + 2p \Rightarrow p = \frac{2}{3}$$

Mixed Nash Equilibrium:

$$\left(\left(\frac{2}{3}, \frac{1}{3} \right), \left(\frac{3}{4}, \frac{1}{4} \right) \right)$$

The Induced joint distribution is:

$$\begin{bmatrix} \frac{2}{3} \cdot \frac{3}{4} & \frac{2}{3} \cdot \frac{1}{4} \\ \frac{1}{3} \cdot \frac{3}{4} & \frac{1}{3} \cdot \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{12} \end{bmatrix}$$

6. A potential game consisting of n players is a normal form game with the property that there exists a function Φ , called a potential function, such that for every $i \in \{1, \dots, n\}$, every $s_i, s'_i \in \mathcal{S}_i$, and every $\mathbf{s}_{-i} \in \mathcal{S}_{-i}$,

$$u_i(s_i, \mathbf{s}_{-i}) - u_i(s'_i, \mathbf{s}_{-i}) = \Phi(s_i, \mathbf{s}_{-i}) - \Phi(s'_i, \mathbf{s}_{-i}) \quad (5)$$

where u_i is the payoff function of player i . Prove that all potential games have a pure-strategy Nash equilibrium.

Remark: Potential games derive their importance from the fact that all *congestion games* have a potential function associated with them. Congestion games model situations wherein a multitude of self-interested agents share common resources. Examples include transportation networks where vehicles share the roads, communication networks where packets share the links, cloud computing where users share clock cycles of the CPU, retail warehousing where sellers share space and manpower of the retail outlets, etc. The above proof implies that every congestion game has a Nash equilibrium in pure strategies.

Solution: Assume finite strategy sets (as in normal-form games discussed in class), so $\mathcal{S} = \prod_{i=1}^n \mathcal{S}_i$ is finite.

Let $\Phi : \mathcal{S} \rightarrow \mathbb{R}$ be an exact potential. Since $\mathcal{S}$ is finite, Φ attains a maximizer; choose

$$s^* \in \arg \max_{s \in \mathcal{S}} \Phi(s).$$

We claim s^* is a pure-strategy Nash equilibrium. Fix player i and any deviation $s'_i \in \mathcal{S}_i$. By maximality of s^* ,

$$\Phi(s_i^*, s_{-i}^*) \geq \Phi(s'_i, s_{-i}^*).$$

Using the potential property,

$$u_i(s_i^*, s_{-i}^*) - u_i(s'_i, s_{-i}^*) = \Phi(s_i^*, s_{-i}^*) - \Phi(s'_i, s_{-i}^*) \geq 0.$$

So $u_i(s_i^*, s_{-i}^*) \geq u_i(s'_i, s_{-i}^*)$ for every s'_i , i.e., player i cannot profitably deviate.

Since this holds for every player i , s^* is a pure Nash equilibrium.

Therefore every potential game has at least one pure-strategy Nash equilibrium.

7. Construct an example and a non-example of a 2-player potential game wherein each player has 4 actions. Prove that the games you construct are indeed potential and non-potential games respectively.

Solution: Let us take an example inspired by the traffic signal problem discussed in class.

$$\mathcal{S}_1 = \mathcal{S}_2 = G_1, G_2, W_1, W_2,$$

where G_k denotes *Go* (with two identical variants) and W_k denotes *Wait*. This simply duplicates actions so that each player has four actions, without changing the strategic structure.

(i) Example of a potential game (traffic intersection).

Consider two drivers arriving at an intersection. Each simultaneously chooses whether to go (G_k) or wait (W_k). Let the total delay experienced by both drivers depend only on whether they go or wait:

$$D(G, G) = 10, \quad D(G, W) = D(W, G) = 2, \quad D(W, W) = 4.$$

(The index k does not affect outcomes.)

Define the potential function

$$\Phi(s) = -D(s),$$

and let each player's payoff be

$$u_1(s) = u_2(s) = \Phi(s).$$

Verification of potential condition:

Fix player 2's action and consider a unilateral deviation of player 1.

For example, when player 2 plays W :

$$u_1(G, W) - u_1(W, W) = (-2) - (-4) = 2,$$

$$\Phi(G, W) - \Phi(W, W) = (-2) - (-4) = 2.$$

Similarly, when player 2 plays G :

$$u_1(G, G) - u_1(W, G) = (-10) - (-2) = -8,$$

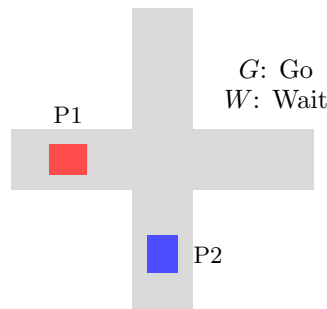
$$\Phi(G, G) - \Phi(W, G) = (-10) - (-2) = -8.$$

Thus,

$$u_1(s_i, s_{-i}) - u_1(s'_i, s_{-i}) = \Phi(s_i, s_{-i}) - \Phi(s'_i, s_{-i}).$$

The same holds for player 2. Hence this is an exact potential game.

Diagram (intersection interpretation):



Each player's payoff aligns with minimizing total system delay. Hence all unilateral incentives are captured by the single function Φ .

(ii) Non-example (not a potential game).

Consider a 2-player zero-sum game with action sets

$$\mathcal{S}_1 = \mathcal{S}_2 = \{A, B, C, D\}.$$

The payoff matrix for player 1 is defined cyclically as follows:

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	0	-1	0	1
<i>B</i>	1	0	-1	0
<i>C</i>	0	1	0	-1
<i>D</i>	-1	0	1	0

and player 2 receives the negative of these payoffs.

This game generalizes the cyclic dominance structure of rock-paper-scissors: each action beats one action and loses to another, creating a cycle of incentives.

Consider the following cycle of unilateral deviations:

$$(A, A) \rightarrow (B, A) \rightarrow (B, B) \rightarrow (C, B) \rightarrow (C, C) \rightarrow (D, C) \rightarrow (D, D) \rightarrow (A, D) \rightarrow (A, A).$$

At each step, exactly one player deviates and strictly improves their payoff. We can verify that each step in the cycle is a unilateral strict improvement as:

- $(A, A) \rightarrow (B, A)$: Player 1 deviates, payoff changes $0 \rightarrow 1$.
- $(B, A) \rightarrow (B, B)$: Player 2 deviates, payoff changes $-1 \rightarrow 0$.
- $(B, B) \rightarrow (C, B)$: Player 1 deviates, payoff changes $0 \rightarrow 1$.
- $(C, B) \rightarrow (C, C)$: Player 2 deviates, payoff changes $-1 \rightarrow 0$.
- $(C, C) \rightarrow (D, C)$: Player 1 deviates, payoff changes $0 \rightarrow 1$.
- $(D, C) \rightarrow (D, D)$: Player 2 deviates, payoff changes $-1 \rightarrow 0$.
- $(D, D) \rightarrow (A, D)$: Player 1 deviates, payoff changes $0 \rightarrow 1$.
- $(A, D) \rightarrow (A, A)$: Player 2 deviates, payoff changes $-1 \rightarrow 0$.

Assume, for contradiction, that a potential function Φ exists. Then every unilateral improvement must strictly increase Φ . Hence, along the above cycle, Φ would strictly increase at every step. However, the cycle returns to the starting profile, which is impossible.

This contradiction implies that no potential function exists.

Hence, this game is not a potential game.

8. This question pertains to the comments in Lecture 6, Slide 6. Consider a two-player zero-sum game and denote by $f(\mathbf{x}, \mathbf{y})$ the expected payoff of the maximizer when the maximizer employs the mixed strategy $\mathbf{x}$ and the minimizer employs the mixed strategy $\mathbf{y}$. Let $(\mathbf{x}^*, \mathbf{y}^*)$ be a Nash equilibrium of the game.

- (a) Show that by employing the strategy $\mathbf{x}^*$, the maximizer is assured an expected payoff of at least $f(\mathbf{x}^*, \mathbf{y}^*)$ regardless of the strategy that the minimizer employs.
- (b) Show that by employing the strategy $\mathbf{y}^*$, the minimizer can keep the maximizer from obtaining an expected payoff of more than $f(\mathbf{x}^*, \mathbf{y}^*)$ regardless of the strategy that the maximizer employs.

Remark: Blackwell Approachability — a celebrated idea in Game theory with applications in many other fields — is a generalization of the above result for vector-valued payoffs.

Solution: Since $(\mathbf{x}^*, \mathbf{y}^*)$ is a Nash equilibrium in a two-player zero-sum game with maximizer payoff f , we have by definition:

$$f(\mathbf{x}^*, \mathbf{y}^*) \geq f(\mathbf{x}, \mathbf{y}^*) \quad \forall \mathbf{x}$$

and

$$f(\mathbf{x}^*, \mathbf{y}^*) \leq f(\mathbf{x}^*, \mathbf{y}) \quad \forall \mathbf{y}.$$

- (a) The second inequality gives

$$f(\mathbf{x}^*, \mathbf{y}) \geq f(\mathbf{x}^*, \mathbf{y}^*) \quad \forall \mathbf{y}.$$

Thus if the maximizer commits to the mixed strategy $\mathbf{x}^*$, the realized expected payoff is never below the equilibrium value $f(\mathbf{x}^*, \mathbf{y}^*)$, regardless of the strategy chosen by the minimizer.

- (b) The first inequality gives

$$f(\mathbf{x}, \mathbf{y}^*) \leq f(\mathbf{x}^*, \mathbf{y}^*) \quad \forall \mathbf{x}.$$

Thus if the minimizer commits to the mixed strategy $\mathbf{y}^*$, the maximizer's payoff can never exceed the equilibrium value, regardless of maximizer strategy.

Therefore $f(\mathbf{x}^*, \mathbf{y}^*)$ is both a guaranteed lower bound for the maximizer via $\mathbf{x}^*$ and a guaranteed upper bound enforced by the minimizer via $\mathbf{y}^*$.

9. Compute the Nash equilibrium of the two-player zero-sum game with the maximizer's payoff matrix

$$\begin{bmatrix} 2 & -4 & 7 \\ -1 & 3 & -4 \\ 5 & -8 & 2 \\ -4 & 5 & 1 \end{bmatrix}. \quad (6)$$

Compute the value of the game.

You can use Matlab (or any other programming language of your choice) to solve Linear Programs.

Solution:

This game can be formulated as a linear program where the row player maximizes a guaranteed payoff v against all column strategies. The constraints ensure that the expected payoff against each pure column strategy is at least v .

$$\text{Payoff matrix: } A = \begin{bmatrix} 2 & -4 & 7 \\ -1 & 3 & -4 \\ 5 & -8 & 2 \\ -4 & 5 & 1 \end{bmatrix}$$

Formulation:

Let $x = (x_1, x_2, x_3, x_4)$ be the mixed strategy of the row player. We maximize the guaranteed payoff v :

$$\max v$$

s.t.

$$\begin{aligned} 2x_1 - x_2 + 5x_3 - 4x_4 &\geq v \\ -4x_1 + 3x_2 - 8x_3 + 5x_4 &\geq v \\ 7x_1 - 4x_2 + 2x_3 + x_4 &\geq v \\ x_1 + x_2 + x_3 + x_4 &= 1 \\ x_i &\geq 0 \end{aligned}$$

Solution (using MATLAB):

$$x^* = (0.375, 0.6103, 0.0147, 0), \quad v \approx 0.2132$$

For verification we can substitute x^* into each column payoff:

$$\text{Col 1} \approx 0.2132$$

$$\text{Col 2} \approx 0.2133$$

$$\text{Col 3} \approx 0.2132$$

All columns yield the same payoff, so the column player is indifferent. Hence, x^* is a Nash equilibrium strategy and v is the value of the game.

Minimizer's strategy:

Let $y = (y_1, y_2, y_3)$ be the mixed strategy of the column player. At equilibrium, the row player must be indifferent among all pure strategies in the support of x^* . Since $x_1, x_2, x_3 > 0$, we impose:

$$\text{Row 1: } 2y_1 - 4y_2 + 7y_3 = v,$$

$$\text{Row 2: } -y_1 + 3y_2 - 4y_3 = v,$$

$$\text{Row 3: } 5y_1 - 8y_2 + 2y_3 = v,$$

with

$$y_1 + y_2 + y_3 = 1, \quad y_j \geq 0.$$

Solving, we obtain

$$y^* \approx (0.5809, 0.3529, 0.0661)$$

Thus, the Nash equilibrium is

$$(x^*, y^*) = ((0.375, 0.6103, 0.0147, 0), (0.5809, 0.3529, 0.0661)),$$

with value

$$v \approx 0.2132.$$

10. The proof of existence of Nash equilibrium that we saw in class has a couple of minor gaps, which you will fill in this question. The notation used in this question is the same as in the lecture slides.
- (a) Show rigorously that the set $\mathcal{X}$ is nonempty, compact, and convex.
 - (b) Show that the expected utility function μ_i of any player i is continuous. Where in the proof is the continuity of μ_i s used?
 - (c) Let $\mathcal{P}_j \subseteq \mathbb{R}^{m_j}$, $j = 1, \dots, n$, be convex sets. Show that $\mathcal{P}_1 \times \dots \times \mathcal{P}_n$ is a convex subset of $\mathbb{R}^{m_1} \times \dots \times \mathbb{R}^{m_n}$. Where in the proof do we use this fact?

Solution:

- (a) For each player i , let $m_i = |\mathcal{S}_i|$ be the number of pure strategies of player i . The mixed-strategy set is

$$\Delta(\mathcal{S}_i) = \left\{ x_i = (x_{i1}, \dots, x_{im_i}) \in \mathbb{R}^{m_i} : x_{ik} \geq 0 \ \forall k, \quad \sum_{k=1}^{m_i} x_{ik} = 1 \right\}.$$

We first show that $\Delta(\mathcal{S}_i)$ is nonempty, compact, and convex.

Non-emptiness:

Consider

$$x_i = \left(\frac{1}{m_i}, \dots, \frac{1}{m_i} \right).$$

Since each coordinate is nonnegative and

$$\sum_{k=1}^{m_i} \frac{1}{m_i} = 1,$$

we have $x_i \in \Delta(\mathcal{S}_i)$. Hence $\Delta(\mathcal{S}_i) \neq \emptyset$.

Compactness:

For every $x_i \in \Delta(\mathcal{S}_i)$,

$$0 \leq x_{ik} \leq 1 \quad \forall k,$$

so $\Delta(\mathcal{S}_i)$ is bounded.

Also,

$$\Delta(\mathcal{S}_i) = \left\{ x_i : \sum_{k=1}^{m_i} x_{ik} = 1 \right\} \cap \bigcap_{k=1}^{m_i} \{x_i : x_{ik} \geq 0\}.$$

The first set is closed since it is the preimage of $\{1\}$ under the continuous map

$$x_i \mapsto \sum_{k=1}^{m_i} x_{ik},$$

and each set $\{x_i : x_{ik} \geq 0\}$ is closed. Hence $\Delta(\mathcal{S}_i)$ is closed.

Therefore $\Delta(\mathcal{S}_i)$ is closed and bounded in $\mathbb{R}^{m_i}$, so by Heine–Borel it is compact.

Convexity:

Let $x_i, y_i \in \Delta(\mathcal{S}_i)$ and let $\lambda \in [0, 1]$. Define

$$z_i = \lambda x_i + (1 - \lambda)y_i.$$

Since $x_{ik}, y_{ik} \geq 0$,

$$z_{ik} = \lambda x_{ik} + (1 - \lambda)y_{ik} \geq 0.$$

Also,

$$\sum_{k=1}^{m_i} z_{ik} = \lambda \sum_{k=1}^{m_i} x_{ik} + (1 - \lambda) \sum_{k=1}^{m_i} y_{ik} = \lambda + (1 - \lambda) = 1.$$

Thus $z_i \in \Delta(\mathcal{S}_i)$, proving convexity.

Now define

$$\mathcal{X} := \prod_{i=1}^n \Delta(\mathcal{S}_i).$$

Non-emptiness of $\mathcal{X}$:

Since each $\Delta(\mathcal{S}_i)$ is nonempty, choose $x_i \in \Delta(\mathcal{S}_i)$ for every i . Then

$$(x_1, \dots, x_n) \in \mathcal{X},$$

so $\mathcal{X} \neq \emptyset$.

Compactness of $\mathcal{X}$:

Each $\Delta(\mathcal{S}_i)$ is compact. Since a finite Cartesian product of compact sets is compact,

$$\mathcal{X} = \prod_{i=1}^n \Delta(\mathcal{S}_i)$$

is compact.

Convexity of $\mathcal{X}$:

Let

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n) \in \mathcal{X},$$

and let $\lambda \in [0, 1]$. Then

$$\lambda x + (1 - \lambda)y = (\lambda x_1 + (1 - \lambda)y_1, \dots, \lambda x_n + (1 - \lambda)y_n).$$

Since each $\Delta(\mathcal{S}_i)$ is convex,

$$\lambda x_i + (1 - \lambda)y_i \in \Delta(\mathcal{S}_i) \quad \forall i.$$

Hence

$$\lambda x + (1 - \lambda)y \in \mathcal{X}.$$

Therefore $\mathcal{X}$ is convex.

(b) Expected utility of player i is

$$\mu_i(x) = \sum_{s \in \mathcal{S}} u_i(s) \prod_{j=1}^n x_j(s_j).$$

This is a finite polynomial in coordinates of x , so it is continuous on $\mathcal{X}$.

Continuity is used when proving the best-response correspondence

$$B_i(x_{-i}) = \arg \max_{x_i \in \Delta(\mathcal{S}_i)} \mu_i(x_i, x_{-i})$$

is nonempty/compact (maximum theorem); these are required to apply Kakutani to $B(x) = \prod_i B_i(x_{-i})$.

(c) Let $p = (p_1, \dots, p_n), q = (q_1, \dots, q_n) \in \mathcal{P}_1 \times \dots \times \mathcal{P}_n$ and $\lambda \in [0, 1]$. Then

$$\lambda p + (1 - \lambda)q = (\lambda p_1 + (1 - \lambda)q_1, \dots, \lambda p_n + (1 - \lambda)q_n).$$

Since each $\mathcal{P}_j$ is convex, $\lambda p_j + (1 - \lambda)q_j \in \mathcal{P}_j$ for every j ; hence

$$\lambda p + (1 - \lambda)q \in \mathcal{P}_1 \times \dots \times \mathcal{P}_n.$$

So the Cartesian product is convex.

In the Nash-existence proof, this is used to conclude that

$$\mathcal{X} = \prod_i \Delta(\mathcal{S}_i)$$

and the value set $B(x) = \prod_i B_i(x_{-i})$ are convex, satisfying Kakutani's convex-valued condition.

Kakutani Fixed Point Theorem:

- ✓ Strategy space X is nonempty, compact, and convex.
- ✓ Best response correspondence $B(x)$ is nonempty for every x .
- ✓ $B(x)$ is convex-valued for every x .
- ✓ $B(x)$ has a closed graph.

11. Prove that every Nash Equilibrium is a Correlated Equilibrium, and that every Correlated Equilibrium is a Coarse Correlated Equilibrium.

Solution:

- (a) Every Nash equilibrium is a correlated equilibrium.

Let $(x_1^*, \dots, x_n^*)$ be a Nash equilibrium. Define the joint distribution

$$p(s) = \prod_{i=1}^n x_i^*(s_i), \quad s = (s_1, \dots, s_n) \in \mathcal{S}.$$

Thus the players' actions are independent under p .

Fix a player i , and let $s_i, s'_i \in S_i$ with

$$p(S_i = s_i) > 0.$$

Since p is a product distribution, conditioning on $S_i = s_i$ does not change the distribution of S_{-i} . Hence

$$p(S_{-i} = s_{-i} \mid S_i = s_i) = \prod_{j \neq i} x_j^*(s_j).$$

Therefore,

$$\mathbb{E}[u_i(s_i, S_{-i}) \mid S_i = s_i] = \sum_{s_{-i}} u_i(s_i, s_{-i}) \prod_{j \neq i} x_j^*(s_j),$$

and similarly,

$$\mathbb{E}[u_i(s'_i, S_{-i}) \mid S_i = s_i] = \sum_{s_{-i}} u_i(s'_i, s_{-i}) \prod_{j \neq i} x_j^*(s_j).$$

Since $p(S_i = s_i) > 0$, we have $x_i^*(s_i) > 0$, so s_i lies in the support of x_i^* . Since $(x_1^*, \dots, x_n^*)$ is a Nash equilibrium, every pure strategy in the support of x_i^* is a best response to x_{-i}^* . Because

$$p(S_i = s_i) > 0,$$

we have $x_i^*(s_i) > 0$, so

$$\sum_{s_{-i}} u_i(s_i, s_{-i}) \prod_{j \neq i} x_j^*(s_j) \geq \sum_{s_{-i}} u_i(s'_i, s_{-i}) \prod_{j \neq i} x_j^*(s_j).$$

Hence,

$$\mathbb{E}[u_i(s_i, S_{-i}) \mid S_i = s_i] \geq \mathbb{E}[u_i(s'_i, S_{-i}) \mid S_i = s_i].$$

Since this holds for every player i , every s_i with positive probability, and every deviation s'_i , the distribution p is a correlated equilibrium.

- (b) Every correlated equilibrium is a coarse correlated equilibrium.

Let p be a correlated equilibrium. Then for every player i , every $s_i \in S_i$ with $p(S_i = s_i) > 0$, and every deviation $s'_i \in S_i$,

$$\mathbb{E}[u_i(s_i, S_{-i}) \mid S_i = s_i] \geq \mathbb{E}[u_i(s'_i, S_{-i}) \mid S_i = s_i].$$

Multiply both sides by $p(S_i = s_i)$ and sum over all s_i :

$$\sum_{s_i} p(S_i = s_i) \mathbb{E}[u_i(s_i, S_{-i}) \mid S_i = s_i] \geq \sum_{s_i} p(S_i = s_i) \mathbb{E}[u_i(s'_i, S_{-i}) \mid S_i = s_i].$$

By the law of total expectation,

$$\sum_{s_i} p(S_i = s_i) \mathbb{E}[u_i(s_i, S_{-i}) \mid S_i = s_i] = \mathbb{E}[u_i(S_i, S_{-i})].$$

Similarly,

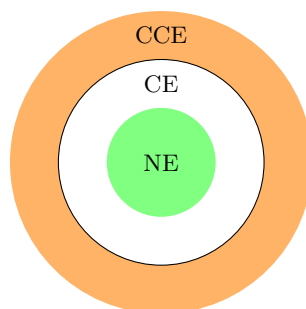
$$\sum_{s_i} p(S_i = s_i) \mathbb{E}[u_i(s'_i, S_{-i}) \mid S_i = s_i] = \mathbb{E}[u_i(s'_i, S_{-i})].$$

Therefore,

$$\mathbb{E}[u_i(S_i, S_{-i})] \geq \mathbb{E}[u_i(s'_i, S_{-i})].$$

Since this holds for every player i and every fixed deviation s'_i , the distribution p is a coarse correlated equilibrium.

We can summarise both results neatly as a venn diagram,



12. The definition of correlated equilibrium that we saw in class states that no player can benefit by unilaterally deviating from the coordinator's recommendation, where the deviation can be to any pure strategy. However, employing pure strategies is not the most general method of play. A player could in general mix over their pure strategies. The goal of this problem is to show that if players cannot improve their payoff by unilaterally deviating to any pure strategy, then they cannot improve their payoff by unilaterally deviating to any mixed strategy either. Consequently, the definition that we saw in class is general enough to encapsulate deviations to mixed strategies.

Consider an n -player game and let $(\hat{S}_1, \dots, \hat{S}_n) \sim p_{CE}$ where p_{CE} is a correlated equilibrium. Show that for every $i \in \{1, \dots, n\}$, every $\hat{s}_i \in \mathcal{S}_i$, and every $S'_i \sim q(\hat{s}_i)$ where $q(\hat{s}_i) \in \Delta(\mathcal{S}_i)$ is any mixed strategy,

$$\mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \geq \mathbb{E}[u_i(S'_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Solution:

Fix a player $i \in \{1, \dots, n\}$ and a recommendation

$$\hat{s}_i \in \mathcal{S}_i$$

such that

$$\Pr(\hat{S}_i = \hat{s}_i) > 0.$$

Define

$$\pi_i(s_{-i} | \hat{s}_i) := \Pr(\hat{S}_{-i} = s_{-i} | \hat{S}_i = \hat{s}_i).$$

Since p_{CE} is a correlated equilibrium, for every pure strategy

$$a \in \mathcal{S}_i,$$

we have

$$\sum_{s_{-i}} \pi_i(s_{-i} | \hat{s}_i) (u_i(\hat{s}_i, s_{-i}) - u_i(a, s_{-i})) \geq 0.$$

Equivalently,

$$\mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \geq \mathbb{E}[u_i(a, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \quad \forall a \in \mathcal{S}_i.$$

Now let

$$S'_i \sim q(\hat{s}_i), \quad q(\hat{s}_i) \in \Delta(\mathcal{S}_i),$$

be any mixed deviation after receiving recommendation $\hat{s}_i$.

Using linearity of expectation,

$$\mathbb{E}[u_i(S'_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] = \sum_{a \in \mathcal{S}_i} q(\hat{s}_i)(a) \mathbb{E}[u_i(a, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

From the correlated equilibrium condition, for every $a \in \mathcal{S}_i$ and every $\hat{s}_i$ with $\Pr(\hat{S}_i = \hat{s}_i) > 0$,

$$\mathbb{E}[u_i(a, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \leq \mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Multiplying each inequality by $q(\hat{s}_i)(a) \geq 0$ and summing over all $a \in \mathcal{S}_i$, we obtain

$$\sum_{a \in \mathcal{S}_i} q(\hat{s}_i)(a) \mathbb{E}[u_i(a, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \leq \sum_{a \in \mathcal{S}_i} q(\hat{s}_i)(a) \mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Since

$$\sum_{a \in \mathcal{S}_i} q(\hat{s}_i)(a) = 1,$$

the right-hand side simplifies to

$$\mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Therefore,

$$\mathbb{E}[u_i(S'_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \leq \mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Hence,

$$\mathbb{E}[u_i(\hat{S}_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i] \geq \mathbb{E}[u_i(S'_i, \hat{S}_{-i}) | \hat{S}_i = \hat{s}_i].$$

Thus, if no unilateral deviation to a pure strategy is profitable, then no unilateral deviation to a mixed strategy is profitable either.